

Erasure Is Deformation

Repair-Repertoire Reduction and the Geometry of Memory Loss

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Abstract

Memory is conventionally modelled as stored information. This paper proposes a different account: memory is repair capacity, and forgetting is the geometric deformation that results when repair operators are removed. We formalise erasure as reduction of a repair repertoire $\mathcal{R}_0 \rightarrow \mathcal{R}_t \subseteq \mathcal{R}_0$ and study its consequences for the observer-relative admissibility manifold $\mathcal{A}(\mathcal{R})$. The Admissibility Contraction Theorem establishes that $\mathcal{A}_t \subseteq \mathcal{A}_0$ with strict inclusion whenever distinctions are lost. The Erasure Deformation Theorem introduces the pullback deformation tensor $E_t = \iota_t^* g_0 - g_t$, where $\iota_t : \text{Rest}(\mathcal{R}_t) \hookrightarrow \text{Rest}(\mathcal{R}_0)$ is the inclusion of surviving repair domains, and proves that $|E_t| \rightarrow \infty$ along sequences approaching the irrestorability boundary. Erasure therefore generates geometric singularities, not merely information gaps. The hysteresis functional $H(d) = C_R^{\text{restore}}(d) - C_R^{\text{destroy}}(d)$ characterises irreversibility: $H(d) = \infty$ is equivalent to permanent forgetting. Topological memory is defined via the homology groups $H_*(\mathcal{A}_t)$; component fragmentation, hole creation, and homology collapse correspond to distinct modes of memory loss. Applications are developed for neural injury, archive destruction, language death, institutional collapse, and scientific knowledge loss. The paper is a sequel to *Observability Is Restorability* [1], extending that framework from the question of what is observable to the question of what happens when observability itself degrades.

1. Memory, Erasure, and Deformation

The standard account of memory is a storage account. Memories are records inscribed in a medium, and forgetting is the degradation or loss of those records. On this picture, memory loss is subtraction: what was present is removed.

This paper begins from a different premise, developed formally in the companion paper *Observability Is Restorability* [1]: memory is not storage but repair capacity. To remember a distinction is to possess a repair path by which that distinction can be restored after perturbation. Forgetting is not the removal of a stored item but the loss of the repair operators by which the item could be reconstructed.

If this is correct, then forgetting has geometric consequences that the storage account cannot capture. When repair operators are removed, the set of restorable distinctions contracts, the observer-relative admissibility manifold deforms, and the metric structure on surviving repair domains is distorted. Erasure is not subtraction. It is deformation.

The paper makes this claim precise. We define erasure as reduction of a repair repertoire, study the geometry of the resulting deformation, characterise irreversibility via a hysteresis functional, and analyse the topological consequences of progressive operator loss. The framework applies equally to neural memory, institutional knowledge, scientific technique, linguistic structure, and any other system whose capacity for distinction-restoration can be degraded.

The argument proceeds as follows. Section 2 reviews repair repertoires, restorable distinctions, and the admissibility manifold from [1]. Section 3 defines erasure events, the erasure semigroup, and the lattice of sub-repertoires. Section 4 proves the Admissibility Contraction Theorem. Section 5 introduces the pull-back deformation tensor and proves the Erasure Deformation Theorem. Section 6 develops the hysteresis functional and the permanent forgetting condition. Section 7 defines topological memory via homology groups of the admissibility manifold. Sections 8–10 develop applications. Section 11 draws consequences and identifies open problems.

2. Repair Repertoires and Admissibility

We recall the essential definitions from [1]. A *substrate* $S = (X_S, \mathcal{D}, \mathcal{D})$ consists of a locally compact Hausdorff state space X_S , a perturbation semigroup $\mathcal{D}$, and a closed distinction relation $\mathcal{D} \subseteq X_S \times X_S$. A *repair repertoire* $\mathcal{R} \subseteq C(X_S, X_S)$ is a sequentially compact set of continuous maps equipped with the compact-open topology. The *minimal repair cost* of distinction d under repertoire $\mathcal{R}$ is

$$C_R(d, \mathcal{R}) = \sup_{p \in \mathcal{D}_d} \inf_{\gamma \in \Gamma_d^p(\mathcal{R})} \text{cost}(\gamma),$$

and the set of *finitely restorable distinctions* is

$$\text{Rest}(\mathcal{R}) = \{d \in \mathcal{D} \mid C_R(d, \mathcal{R}) < +\infty\}.$$

The *repair pseudometric* on $\text{Rest}(\mathcal{R})$ is

$$d_R(d_i, d_j) = \inf_{\gamma: d_i \rightsquigarrow d_j} \int_{\gamma} c \, ds,$$

which defines a metric $g_{\mathcal{R}}$ on $\text{Rest}(\mathcal{R})$ in the reversible case. The *observer-relative admissibility manifold* is

$$A(\mathcal{R}) = \overline{\{x \in X_S \mid \forall d \in \text{Rest}(\mathcal{R}), C_R(d, \mathcal{R}) < +\infty \text{ from } x\}}.$$

The Repair Boundary Theorem [1] states that $x \in \partial A(\mathcal{R})$ if and only if $C_R(d, \mathcal{R}) \rightarrow +\infty$ for some $d \in \text{Rest}(\mathcal{R})$ as the system approaches x .

Remark 2.1. Throughout this paper, *memory* denotes $\text{Rest}(\mathcal{R})$ together with its metric structure $g_{\mathcal{R}}$. This is not a storage register but a geometric object: the metrised space of distinctions the observer can maintain. Forgetting is change of this object.

Remark 2.2 (Observer persistence). Throughout this paper we assume that the observer persists across the erasure path: $\mathcal{R}_0 \rightarrow \mathcal{R}_t$ is a reduction belonging to the same observer, not a transition to a different observer. This assumption is non-trivial. Severe forgetting — traumatic brain injury, late-stage neurodegeneration, total institutional collapse — may change not only what the observer can restore but which distinctions the observer constitutes. In those cases $(A(\mathcal{R}_0), g_0)$ and (A_t, g_t) may be geometries of different observers rather than different states of the same one. The question of whether observer identity is itself defined by

a path in the category of admissibility manifolds — so that forgetting is *observer deformation* rather than memory deformation — is left for future work. The current framework is exactly correct for the regime in which identity persists: partial or moderate erasure where the surviving repair domain is non-empty and the observer remains recognisably continuous.

Remark 2.3 (The invariant making comparison meaningful). The reviewer of an earlier draft raised the question: if both the observer and the distinction set change under erasure, what invariant makes the comparison $E_t = \iota_t^* g_0 - g_t$ meaningful? The answer is the inclusion $\iota_t : \text{Rest}(\mathcal{R}_t) \hookrightarrow \text{Rest}(\mathcal{R}_0)$. The pullback construction exists precisely because surviving distinctions are identified with their original representatives via ι_t . The deformation tensor measures how the geometry of those surviving distinctions has changed, relative to the geometry they had before erasure. No universal distinction space $\mathcal{D}_\infty$ is needed and none is assumed; the entire framework is observer-relative, which is the correct register for a repair-theoretic account. Introducing a universal distinction space would move the theory toward latent-space realism, which the parent paper [1] was designed to avoid.

3. Erasure: Operators, Events, and the Sub-Repertoire Lattice

Definition 3.1 (Erasure event). An *erasure event* at time t is a map $\varepsilon_t : \mathcal{R} \mapsto \mathcal{R}'$ with $\mathcal{R}' \subseteq \mathcal{R}$, removing one or more repair operators from the available repertoire. An erasure event is *local* if it removes operators acting on a bounded region of X_S ; *global* if it removes operators whose domain is all of X_S .

Definition 3.2 (Erasure path). An *erasure path* is a decreasing family of repertoires $(\mathcal{R}_t)_{t \in [0, T]}$ with $\mathcal{R}_s \subseteq \mathcal{R}_t$ for all $s \geq t$, and $\mathcal{R}_0$ the initial repertoire. The family is parameterised by t increasing in the direction of greater loss.

Definition 3.3 (Erasure monoid and lattice action). The set of all erasure events under composition forms a *monoid* $(\mathcal{E}, \circ, \varepsilon_\emptyset)$: the composition $\varepsilon_s \circ \varepsilon_t$ applies ε_t first, then ε_s ; the identity is the null erasure $\varepsilon_\emptyset = \text{id}$. The monoid is not in general a group: erasure events need not be invertible.

The monoid $\mathcal{E}$ acts on the sub-repertoire lattice $\mathcal{L}(\mathcal{R}_0)$ by

$$\mathcal{E} \times \mathcal{L}(\mathcal{R}_0) \longrightarrow \mathcal{L}(\mathcal{R}_0), \quad (\varepsilon, \mathcal{R}) \mapsto \varepsilon(\mathcal{R}) \subseteq \mathcal{R}.$$

An erasure path is an orbit segment of this action. Path-dependence of erasure is the statement that two orbits with the same endpoints need not traverse the same intermediate lattice elements, producing different intermediate admissibility geometries. The orbit structure of $\mathcal{E}$ on $\mathcal{L}(\mathcal{R}_0)$ encodes all possible forgetting trajectories from $\mathcal{R}_0$.

Remark 3.4 (Structure over elements). Meadows [12] argues that in any complex system, the interconnections and feedback loops among elements are more determinative of system behaviour than the elements themselves. The repair-theoretic account instantiates this principle precisely: the admissibility geometry $\mathcal{A}(\mathcal{R})$ and the deformation tensor E_t are properties of the *structure* of the repair repertoire — how operators compose, what they jointly restore, how their removal cascades through the lattice $\mathcal{L}(\mathcal{R}_0)$ — not properties of individual operators in isolation. Erasing one operator changes the geometry not only of the distinctions that operator directly maintained but, through the causal linkages encoded in the lattice, of every distinction whose repair paths passed through it. This is the formal analogue of Meadows’ observation that changing a single connection in a system can alter the entire pattern of its behaviour.

Definition 3.5 (Sub-repertoire lattice). The set of all sub-repertoires of $\mathcal{R}_0$, ordered by inclusion, forms a complete lattice $(\mathcal{L}(\mathcal{R}_0), \subseteq)$. Meets are intersections; joins are unions (when the union is still a valid repertoire). An erasure path is a descending chain in this lattice.

Remark 3.6 (Path-dependence of erasure). Two erasure paths with the same endpoints $\mathcal{R}_0 \rightarrow \mathcal{R}_T$ may pass through different intermediate repertoires and produce different intermediate admissibility geometries, even when $\mathcal{A}(\mathcal{R}_T)$ is identical. This path-dependence is the first indication that erasure is a geometric process, not merely a set-theoretic one. It is the precursor of the hysteresis phenomenon studied in Section 6.

Definition 3.7 (Partial and total repertoire collapse). An erasure path reaches *partial collapse* at time t if $\text{Rest}(\mathcal{R}_t) \subsetneq \text{Rest}(\mathcal{R}_0)$ with $\text{Rest}(\mathcal{R}_t) \neq \emptyset$. It reaches *total collapse* if $\text{Rest}(\mathcal{R}_t) = \emptyset$. Total collapse corresponds to the observer losing all observational capacity.

4. Admissibility Contraction

Theorem 4.1 (Admissibility Contraction Theorem). *Let $\mathcal{R}_t \subseteq \mathcal{R}_0$ be a repair-repertoire reduction. Then*

$$A(\mathcal{R}_t) \subseteq A(\mathcal{R}_0).$$

Moreover, if $\text{Rest}(\mathcal{R}_t) \subsetneq \text{Rest}(\mathcal{R}_0)$, the inclusion is strict and at least one boundary component of $A(\mathcal{R}_0)$ migrates inward: there exists a point $x \in A(\mathcal{R}_0)$ such that $x \notin A(\mathcal{R}_t)$.

Proof. ($\subseteq$) Suppose $x \in A(\mathcal{R}_t)$. Then from x , every distinction in $\text{Rest}(\mathcal{R}_t)$ is finitely restorable under $\mathcal{R}_t$. Since $\mathcal{R}_t \subseteq \mathcal{R}_0$, every repair path available in $\mathcal{R}_t$ is also available in $\mathcal{R}_0$. Hence every distinction in $\text{Rest}(\mathcal{R}_t) \subseteq \text{Rest}(\mathcal{R}_0)$ is also finitely restorable under $\mathcal{R}_0$ from x , so $x \in A(\mathcal{R}_0)$.

Strict inclusion. Suppose $d^* \in \text{Rest}(\mathcal{R}_0) \setminus \text{Rest}(\mathcal{R}_t)$. Then $C_R(d^*, \mathcal{R}_t) = +\infty$: there exists an admissible perturbation of d^* from which no finite repair path in $\mathcal{R}_t$ restores it. Let x_0 be a state from which d^* was restorable under $\mathcal{R}_0$. Since $d^* \notin \text{Rest}(\mathcal{R}_t)$, there is no finite repair path from x_0 under $\mathcal{R}_t$ restoring d^* . Therefore $x_0 \notin A(\mathcal{R}_t)$. But $x_0 \in A(\mathcal{R}_0)$, establishing strict inclusion. By continuity of $C_R(\cdot, \mathcal{R}_0)$, the set of such points x_0 is non-empty and open, so a boundary component of $A(\mathcal{R}_0)$ has migrated inward. $\square$

Corollary 4.2 (Monotonicity of memory under erasure). *Repair-repertoire reduction monotonically contracts memory:*

$$\mathcal{R}_t \subseteq \mathcal{R}_0 \implies \text{Rest}(\mathcal{R}_t) \subseteq \text{Rest}(\mathcal{R}_0).$$

This is the direct analogue of the Repair Monotonicity Theorem of [1], now stated in the erasure direction.

Corollary 4.3 (Boundary migration). *Under a non-trivial erasure path $(\mathcal{R}_t)_{t \in [0, T]}$, the admissibility boundary $\partial A(\mathcal{R}_t)$ migrates inward monotonically: $\partial A(\mathcal{R}_s)$ is strictly interior to $A(\mathcal{R}_t)$ for $s > t$ whenever the erasure removes distinctions. The observer's world contracts with every operator removed.*

5. The Erasure Deformation Tensor

The Admissibility Contraction Theorem is a set-theoretic result. It tells us that the repair domain shrinks but not how it is distorted. To capture distortion we need a tensor.

The key point is that after erasure, the surviving repair domain $\text{Rest}(\mathcal{R}_t)$ sits inside the original domain $\text{Rest}(\mathcal{R}_0)$ via the inclusion

$$\iota_t : \text{Rest}(\mathcal{R}_t) \hookrightarrow \text{Rest}(\mathcal{R}_0).$$

The original metric $g_0 = g_{\mathcal{R}_0}$ pulls back to $\iota_t^* g_0$ on $\text{Rest}(\mathcal{R}_t)$. The reduced repertoire $\mathcal{R}_t$ induces its own metric $g_t = g_{\mathcal{R}_t}$ on $\text{Rest}(\mathcal{R}_t)$. These need not agree.

Definition 5.1 (Erasure deformation tensor). The *erasure deformation tensor* at time t is

$$E_t = \iota_t^* g_0 - g_t,$$

a symmetric $(0,2)$ -tensor on $\text{Rest}(\mathcal{R}_t)$. Positive definiteness of E_t at a distinction d means the original metric assigns greater distances near d than the reduced metric; negative definiteness means the reduced metric is locally more spread.

Remark 5.2. The pullback construction is essential. Writing $E_t = g_0 - g_t$ would compare metrics on different spaces, since g_0 is defined on $\text{Rest}(\mathcal{R}_0)$ and g_t is defined on the strictly smaller set $\text{Rest}(\mathcal{R}_t)$. The inclusion ι_t identifies surviving distinctions with their images in the original space, so the comparison is meaningful: E_t measures how erasure has distorted the geometry of what remains, measured against the geometry it formerly had.

Lemma 5.3 (Repair Cost Coercivity). Let $d_R^{(t)}$ denote the repair pseudometric induced by the repertoire $\mathcal{R}_t$. Assume that for every $M < +\infty$, the set of repair paths of cost at most M under $\mathcal{R}_t$,

$$\Gamma_M^{(t)} = \{\gamma \in \mathcal{R}_t^* \mid \text{cost}(\gamma) \leq M\},$$

is sequentially compact under the product topology induced by the compact-open topology on $\mathcal{R}_t$. Suppose $d^* \in \text{Rest}(\mathcal{R}_0) \setminus \text{Rest}(\mathcal{R}_t)$ and $(d_n)_{n \geq 1}$ is a sequence in $\text{Rest}(\mathcal{R}_t)$ with $d_n \rightarrow d^*$ in the g_0 -topology. Then for every fixed $d_0 \in \text{Rest}(\mathcal{R}_t)$ not approaching d^* ,

$$d_R^{(t)}(d_n, d_0) \rightarrow +\infty.$$

Proof. Suppose for contradiction that the distances remain bounded: $d_R^{(t)}(d_n, d_0) \leq M$ for all sufficiently large n and some $M < +\infty$. Then there exists a sequence of repair paths $(\gamma_n) \subseteq \Gamma_{M+1}^{(t)}$ with $\gamma_n : d_n \rightsquigarrow d_0$.

By the assumed sequential compactness of $\Gamma_{M+1}^{(t)}$, a subsequence (γ_{n_k}) converges to a limit path $\gamma^* \in \Gamma_{M+1}^{(t)}$. By joint continuity of evaluation in the compact-open topology on the locally compact Hausdorff substrate X_S (Axioms 1–3 of [1]), the terminal states converge: $d_{n_k} \rightarrow d^*$ implies γ^* connects d^* to d_0 . Since the distinction relation $\mathcal{D}$ is closed (Axiom 2 of [1]) and the terminal state of γ_{n_k} lies in $\mathcal{D}$, the terminal state of γ^* also lies in $\mathcal{D}$. By lower semi-continuity of the cost functional,

$$\text{cost}(\gamma^*) \leq \liminf_k \text{cost}(\gamma_{n_k}) \leq M + 1 < +\infty.$$

This means d^* is finitely restorable under $\mathcal{R}_t$, contradicting $d^* \notin \text{Rest}(\mathcal{R}_t)$. Therefore no uniform bound M exists, and $d_R^{(t)}(d_n, d_0) \rightarrow +\infty$. $\square$

Remark 5.4 (On the path-space compactness hypothesis). The hypothesis that $\Gamma_M^{(t)}$ is sequentially compact is the path-space analogue of Axiom 3 (compact repair repertoire) from [1]. It holds whenever $\mathcal{R}_t$ is a compact set of continuous maps and the set of efficient paths of bounded cost can be uniformly parameterised — for instance, when paths have uniformly bounded length under the cost density c . Without this assumption, a bounded-cost sequence of repair paths can oscillate indefinitely without converging, leaving the contradiction argument incomplete. The assumption is satisfied in all concrete applications considered in this paper.

Theorem 5.5 (Erasure Deformation Theorem). *Let $(\mathcal{R}_t)_{t \geq 0}$ be an erasure path satisfying the path-space compactness hypothesis of Lemma 5.3, and let $(d_n)_{n \geq 1}$ be a sequence in $\text{Rest}(\mathcal{R}_t)$ approaching a distinction $d^* \in \text{Rest}(\mathcal{R}_0) \setminus \text{Rest}(\mathcal{R}_t)$ in the g_0 -topology. Fix any $d_0 \in \text{Rest}(\mathcal{R}_t)$ not approaching d^* , and let $|\cdot|$ denote the operator norm on symmetric $(0, 2)$ -tensors induced by g_0 . Then*

$$|E_t|(d_n) \rightarrow +\infty \quad \text{as } d_n \rightarrow d^*.$$

Erasure generates geometric singularities at the irrestorability boundary.

Proof. We evaluate the two components of $E_t = \iota_t^* g_0 - g(d_R^{(t)})$ along the sequence (d_n) , where $g(d_R^{(t)})$ denotes the $(0, 2)$ -tensor induced by the repair pseudometric

$d_R^{(t)} = d_R^{(t)}$ under the identification of $\text{Rest}(\mathcal{R}_t)$ as a metric space (in the reversible case) or, more generally, the bilinear form $g(d_R^{(t)})(u, v) = d_R^{(t)}(u, v) + d_R^{(t)}(v, u)$ symmetrised from the pseudometric.

Pullback component is bounded. The pullback metric $(\iota_t^* g_0)(d_n, d_0)$ is the g_0 -distance between d_n and d_0 measured in the intact repair space $\text{Rest}(\mathcal{R}_0)$. Since $d_n \rightarrow d^*$ in $\text{Rest}(\mathcal{R}_0)$ and d_0 is fixed away from d^* , the sequence $(\iota_t^* g_0)(d_n, d_0)$ converges to $g_0(d^*, d_0) < +\infty$. In particular, $(\iota_t^* g_0)(d_n, d_0)$ is bounded above.

Induced form from reduced pseudometric diverges. By Lemma 5.3, $d_R^{(t)}(d_n, d_0) \rightarrow +\infty$. Therefore $g(d_R^{(t)})(d_n, d_0) \rightarrow +\infty$.

Tensor norm diverges. The operator norm satisfies

$$|E_t|(d_n) \geq |(\iota_t^* g_0)(d_n, d_0) - g(d_R^{(t)})(d_n, d_0)|.$$

Since $(\iota_t^* g_0)(d_n, d_0)$ is bounded while $g(d_R^{(t)})(d_n, d_0) \rightarrow +\infty$, the right-hand side diverges. Therefore $|E_t|(d_n) \rightarrow +\infty$. $\square$

Remark 5.6. Theorem 5.5 is the geometric content of the title. Erasure does not merely remove distinctions; it deforms the metric structure of surviving distinctions near the lost boundary. An observer approaching the irrestorability boundary experiences not just loss but increasing distortion of what remains. This captures phenomenological features of memory degradation: distinctions near the boundary of recall become harder to separate, more costly to maintain, and eventually indistinguishable.

Proposition 5.7 (Metric degeneration at the irrestorability boundary). *Let g_t be a smooth Riemannian metric on $\text{Rest}(\mathcal{R}_t)$ (in the reversible case, where the repair pseudometric satisfies the symmetry condition of Proposition 3.5 in [1]). If $|E_t|(d_n) \rightarrow +\infty$ along a sequence approaching an irrestorability boundary, then either:*

1. *the smallest eigenvalue of g_t approaches zero (metric collapse: the manifold develops a degenerate direction), or*
2. *the largest eigenvalue of g_t diverges (metric blow-up: distances in some direction become infinite).*

In either case, the metric g_t becomes singular at the irrestorability boundary.

Proof. Since $|E_t| \rightarrow +\infty$ and $\iota_t^* g_0$ is bounded (Theorem 5.5), the eigenvalues of g_t must be unbounded or approach zero along the sequence, for otherwise $|E_t|$ would remain bounded. The two cases correspond to opposite-signed singularities of the metric. $\square$

Corollary 5.8 (Forgetting generates metric singularities). *Under the conditions of Proposition 5.7, curvature divergence follows if g_t has bounded derivatives. Without this regularity assumption, the weaker statement is that the metric becomes singular — which is a necessary condition for curvature blow-up but does not imply it in full generality. Forgetting is therefore curvature change in the regular case, and metric degeneration in general.*

6. Hysteresis and Irreversibility

The deformation tensor describes the geometry of erasure at a fixed time. The hysteresis functional describes the dynamics of restoration after erasure.

Definition 6.1 (Destruction and restoration cost). For a distinction $d \in \text{Rest}(\mathcal{R}_0)$, define:

- the *destruction cost* $C_R^{\text{destroy}}(d)$: the minimal cost required to remove an operator from $\mathcal{R}_0$ such that d becomes irrestorable;
- the *restoration cost* $C_R^{\text{restore}}(d, \mathcal{R}_t)$: the minimal cost of re-acquiring operators sufficient to return d to $\text{Rest}(\mathcal{R}_t)$, starting from the reduced repertoire $\mathcal{R}_t$.

Both costs admit concrete operational interpretations: in biological systems, cost is metabolic energy or cellular resources ($C_R^{\text{destroy}} = \text{energy to eliminate a synaptic pathway}$; $C_R^{\text{restore}} = \text{energy to grow a new one}$); in institutional systems, cost is organisational resources ($C_R^{\text{destroy}} = \text{cost of decommissioning an archive or retiring a specialist}$; $C_R^{\text{restore}} = \text{cost of rebuilding equivalent capacity from scratch}$); in linguistic systems, cost is measurable in speaker-years ($C_R^{\text{destroy}} = \text{attrition required to make a distinction irrestorable}$; $C_R^{\text{restore}} = \text{revitalisation effort required to restore it}$). The key asymmetry is that destruction cost is measured going down the sub-repertoire lattice, while restoration cost is measured going up from $\mathcal{R}_t$; the hysteresis functional captures this asymmetry in a single signed quantity.

Definition 6.2 (Hysteresis functional). The *hysteresis functional* of distinction d under erasure path $(\mathcal{R}_t)$ is

$$H_t(d) = C_R^{\text{restore}}(d, \mathcal{R}_t) - C_R^{\text{destroy}}(d).$$

Three regimes characterise the hysteresis functional:

- $H_t(d) = 0$: *reversible erasure*. Restoration costs exactly as much as destruction. The distinction is forgotten and recovered with equal effort. This regime governs ordinary perturbation within the admissibility interior.
- $0 < H_t(d) < +\infty$: *partial hysteresis*. Restoration is more costly than destruction. This is the generic biological and institutional regime: synaptic weakening is reversible but synaptic elimination requires costly re-growth; institutional procedures once suspended require substantial effort to reinstate.
- $H_t(d) = +\infty$: *permanent forgetting*. The distinction cannot be restored from $\mathcal{R}_t$ at any finite cost.

Theorem 6.3 (Permanent Forgetting Theorem). $H_t(d) = +\infty$ if and only if $d \notin \text{Rest}(\mathcal{R}_t)$. Infinite hysteresis is equivalent to the distinction having crossed the irrestorability boundary.

Proof. ($\Rightarrow$) If $H_t(d) = +\infty$, then $C_R^{\text{restore}}(d, \mathcal{R}_t) = +\infty$: no finite re-acquisition of operators restores d from $\mathcal{R}_t$. This means $d \notin \text{Rest}(\mathcal{R}_t)$.

($\Leftarrow$) If $d \notin \text{Rest}(\mathcal{R}_t)$, then d is not finitely restorable under $\mathcal{R}_t$. Any sequence of operator acquisitions that would restore d must ultimately recover a repertoire $\mathcal{R}'$ with $d \in \text{Rest}(\mathcal{R}')$. The cost of doing so from $\mathcal{R}_t$ is $C_R^{\text{restore}}(d, \mathcal{R}_t) = +\infty$ if no finite extension of $\mathcal{R}_t$ suffices — which holds whenever the operators needed to restore d are themselves irrecoverable (as in biological cell death or archive destruction). Hence $H_t(d) = +\infty$. $\square$

Remark 6.4. The Permanent Forgetting Theorem connects the dynamical notion (infinite restoration cost) to the geometric notion (crossing the admissibility boundary). It is the hysteresis analogue of the Repair Boundary Theorem of

[1]: where that theorem characterised the boundary via diverging repair cost, this theorem characterises it via diverging restoration cost after the boundary has been crossed.

Proposition 6.5 (Monotonicity of hysteresis under progressive erasure). *Along an erasure path $(\mathcal{R}_t)_{t \geq 0}$, the hysteresis functional is non-decreasing in t :*

$$s \geq t \implies H_s(d) \geq H_t(d)$$

for all $d \in \text{Rest}(\mathcal{R}_0)$. Further erasure never makes restoration easier.

Proof. Since $\mathcal{R}_s \subseteq \mathcal{R}_t$ for $s \geq t$, the set of operators available for restoration at time s is a subset of those available at time t . Therefore $C_R^{\text{restore}}(d, \mathcal{R}_s) \geq C_R^{\text{restore}}(d, \mathcal{R}_t)$, giving $H_s(d) \geq H_t(d)$. $\square$

Theorem 6.6 (Hysteresis–Boundary Correspondence). *Let $d \in \text{Rest}(\mathcal{R}_0)$. Then*

$$H_t(d) = +\infty \iff \inf_{\gamma} \left(\sup_{x \in \gamma} C_R(x, \mathcal{R}_t) \right) = +\infty,$$

where the infimum is taken over all restoration trajectories γ from $\mathcal{R}_t$ to any repertoire $\mathcal{R}' \supseteq \mathcal{R}_t$ with $d \in \text{Rest}(\mathcal{R}')$. Permanent forgetting is equivalent to every restoration trajectory encountering a minimax barrier of infinite repair cost.

Proof. $(\implies)$ Suppose $H_t(d) = +\infty$, so $C_R^{\text{restore}}(d, \mathcal{R}_t) = +\infty$. Suppose for contradiction that there exists a restoration trajectory γ^* with

$$\sup_{x \in \gamma^*} C_R(x, \mathcal{R}_t) < +\infty.$$

Then every point along γ^* lies at finite repair cost, and the total restoration cost accumulated along γ^* is bounded above by the path length times this finite supremum. Therefore $C_R^{\text{restore}}(d, \mathcal{R}_t) < +\infty$, contradicting the hypothesis. Hence no such trajectory exists: every restoration trajectory γ satisfies $\sup_{x \in \gamma} C_R(x, \mathcal{R}_t) = +\infty$. Taking the infimum over all trajectories gives $\inf_{\gamma} \sup_{x \in \gamma} C_R(x, \mathcal{R}_t) = +\infty$.

$(\impliedby)$ Suppose $\inf_{\gamma} \sup_{x \in \gamma} C_R(x, \mathcal{R}_t) = +\infty$. Then for every restoration trajectory γ , $\sup_{x \in \gamma} C_R(x, \mathcal{R}_t) = +\infty$. By the Repair Boundary Theorem of [1], points where $C_R(\cdot, \mathcal{R}_t) \rightarrow +\infty$ are precisely points on the irrestorability boundary $\partial A(\mathcal{R}_t)$. Every trajectory therefore accumulates at boundary points of infinite repair cost and cannot reach d at finite total cost. Hence $C_R^{\text{restore}}(d, \mathcal{R}_t) = +\infty$, giving $H_t(d) = +\infty$. $\square$

Remark 6.7. The Hysteresis–Boundary Correspondence gives permanent forgetting a variational characterisation: $H_t(d) = +\infty$ if and only if the minimax repair cost over all restoration trajectories is infinite. This is the exact analogue of barrier-crossing formulations in control theory, large deviations theory, and energy landscape analysis, where a transition between states is impossible if the infimum of the worst-case cost over all paths is infinite. The theorem unifies admissibility geometry (the boundary), dynamical cost (the hysteresis functional), and the Permanent Forgetting Theorem into a single minimax equivalence. Every concept introduced in this paper — contraction, deformation, hysteresis, permanent forgetting — is now a consequence of a single geometric fact: that irrestorability boundaries are minimax barriers of infinite repair cost.

Theorem 6.8 (Boundary Migration). *Let $(\mathcal{R}_t)_{t \geq 0}$ be a non-trivial erasure path. For each $t > 0$ such that $\text{Rest}(\mathcal{R}_t) \subsetneq \text{Rest}(\mathcal{R}_0)$, there exists an open set $U_t \subset A_0$ with $U_t \cap A_t = \emptyset$ and $U_t \cap \partial A_t \neq \emptyset$: the irrestorability boundary ∂A_t has migrated inward into the former interior of A_0 .*

Proof. By the Admissibility Contraction Theorem, $A_t \subseteq A_0$ with strict inclusion when $\text{Rest}(\mathcal{R}_t) \subsetneq \text{Rest}(\mathcal{R}_0)$. Let $x_0 \in A_0 \setminus A_t$. Since A_0 is the closure of its interior, points arbitrarily close to x_0 were formerly in the interior of A_0 . As $A_t \not\ni x_0$, the boundary ∂A_t lies between x_0 and the interior of A_t , establishing U_t . $\square$

Definition 6.9 (Boundary migration measure). The *boundary migration measure* at time t is

$$V(t) = \inf_{x \in \partial A_t} \inf_{y \in \partial A_0} d_R^{(0)}(x, y),$$

where $d_R^{(0)}$ is the original repair pseudometric and the double infimum gives the point-to-set distance from the current irrestorability boundary ∂A_t to the original boundary ∂A_0 in the original repair geometry. $V(t)$ is non-negative with $V(t) > 0$ whenever the boundary has migrated inward. Whether $V(t)$ is differentiable in t , and what the derivative would represent as a migration speed, depends on regularity assumptions about the boundary evolution not established here; this is an open problem.

Remark 6.10. Theorem 6.8 reformulates forgetting as a moving-front problem. The irrestorability boundary advances inward through the admissibility manifold, sweeping formerly admissible states into the irrestorable region. The migration measure $V(t)$ quantifies how far the front has advanced without requir-

ing smoothness of the boundary, making it applicable even when the boundary fragments or develops topological complexity under erasure. Biological analogues include the penumbra in stroke and the retreat of phonological distinctions as speaker populations decline.

7. Topological Memory

The deformation tensor and hysteresis functional are local objects: they describe geometry near individual distinctions and boundaries. To capture global structure of memory loss, we need topology.

The metric and topological levels of the framework answer different questions and should be clearly distinguished. The metric structure — the repair pseudometric $d_R^{(t)}$ and the deformation tensor E_t — answers the question: *how costly is access to a distinction?* It is quantitative, local, and sensitive to the specific distances between distinctions in the repair domain. The topological structure — the homology groups $H_k(A_t)$ and the persistence module $PH_*(A)$ — answers the question: *which distinctions remain mutually accessible at all?* It is qualitative, global, and invariant under homeomorphism. These are independent. Large metric deformation can occur without any topological change (if the repair domain contracts but remains connected and simply connected). Conversely, topological change — fragmentation, hole creation — can occur with bounded local metric deformation. The full geometry of forgetting requires both levels: topology describes the qualitative structure of what remains, metric geometry describes the quantitative structure of how easily it is accessed.

The erasure path $(\mathcal{R}_t)_{t \geq 0}$ generates a filtration of admissibility manifolds:

$$A_0 \supseteq A(\mathcal{R}_1) \supseteq A(\mathcal{R}_2) \supseteq \dots$$

This filtration is the natural input to persistent homology. The individual homology groups $H_k(A_t)$ record topology at a single time t ; the persistence module records how that topology evolves across the entire erasure path.

Definition 7.1 (Persistence module of memory). The *persistence module* of the observer’s memory under the erasure path $(\mathcal{R}_t)$ is the collection of maps

$$\mathbf{PH}_k = (H_k(A_t; \mathbb{Z}), \phi_{s,t} : H_k(A_t) \rightarrow H_k(A(\mathcal{R}_s)))_{s \geq t},$$

where $\phi_{s,t}$ is induced by the inclusion $A(\mathcal{R}_s) \hookrightarrow A_t$. The *persistent topological memory* is

$$PH_*(A) = \bigoplus_{k \geq 0} \mathbf{PH}_k.$$

The persistence diagram of $PH_*(A)$ records, for each homological feature, its birth time t_b (the erasure time at which it first appears) and death time t_d (the time at which it is destroyed by further erasure). Features with large lifespan $t_d - t_b$ represent structurally robust repair capacity; features with small lifespan represent fragile repair cycles sensitive to operator loss.

Remark 7.2 (On homology coefficients). We use $\mathbb{Z}$ coefficients throughout. With $\mathbb{Z}$ coefficients, $H_0(A_t; \mathbb{Z})$ is the free abelian group on connected components of A_t , so $\text{rank}(H_0) = \text{number of components}$ — exactly the quantity relevant for the Fragmentation Theorem. With $\mathbb{Z}/2$ coefficients the rank would agree for orientable manifolds, but A_t need not be orientable after erasure. Integer coefficients are the natural choice when we wish to count components and loops without orientability assumptions, which is the appropriate setting for admissibility manifolds whose topology may be altered by non-orientable erasure events.

Definition 7.3 (Topological memory). The *topological memory* of an observer at time t is the graded object

$$\mathcal{M}_t = \bigoplus_{k \geq 0} H_k(A_t; \mathbb{Z}),$$

where $A_t = A(\mathcal{R}_t)$ is the admissibility manifold at time t . The full memory across the erasure path is $PH_*(A)$.

Each homology group captures a distinct mode of persistent memory structure:

- $H_0(A_t)$: connected components of the admissibility manifold. A single component corresponds to a globally connected repair space; multiple components indicate *repair domain fragmentation* — regions of the distinction space that can no longer be jointly maintained.
- $H_1(A_t)$: loops in the admissibility manifold. Non-trivial loops correspond to *repair cycles* — closed sequences of repair operations that return to their starting point. Their birth and death in PH_1 records the creation and destruction of circular repair redundancy under erasure.

- $H_k(A_t)$ for $k \geq 2$: higher-order topological features corresponding to repair structures with more complex boundary behaviour.

Definition 7.4 (Topological forgetting). A mode of *topological forgetting* occurs when an erasure event changes the homology of A_t :

- *Fragmentation*: $\text{rank}(H_0(A_t))$ increases.
- *Hole creation*: $\text{rank}(H_1(A_t))$ increases.
- *Homology collapse*: $\text{rank}(H_k(A_t))$ decreases for some k .

All three modes are visible in the persistence diagram as births, long-lived features, and deaths respectively.

Theorem 7.5 (Fragmentation Theorem). *Let $(\mathcal{R}_t)_{t \geq 0}$ be an erasure path that removes a distinction $d^* \in \text{Rest}(\mathcal{R}_0)$. If d^* was a cut-point of A_0 , then the erasure event that removes d^* increases $\text{rank}(H_0(A_t))$ by at least one. Fragmentation is the topological signature of the loss of a structurally critical distinction.*

Proof. If d^* is a cut-point of A_0 , then $A_0 \setminus \{d^*\}$ is disconnected. After the erasure event making d^* irrestorable, A_t is contained in $A_0 \setminus \{d^*\}$ and is therefore disconnected, increasing $\text{rank}(H_0(A_t))$. $\square$

Theorem 7.6 (Hole Creation Theorem). *Let $U \subset A_t$ be a contractible region of the admissibility manifold whose repair operators are entirely removed by an erasure event, so that $U \cap A(\mathcal{R}_{t+\varepsilon}) = \emptyset$ for all small $\varepsilon > 0$. If the removal of U does not disconnect A_t but creates a non-contractible loop in $A_t \setminus U$, then*

$$\text{rank}(H_1(A(\mathcal{R}_{t+\varepsilon}))) > \text{rank}(H_1(A_t)).$$

Erasure can increase topological complexity even while decreasing admissible volume.

Proof. The removed region U becomes inaccessible under the reduced repertoire. The complement $A_t \setminus U$ contains a loop surrounding U that is not the boundary of any admissible 2-chain in $A_t \setminus U$: no 2-chain can fill the loop because U is no longer part of the admissible space. Hence a new non-trivial homology class appears in H_1 , and the rank increases. $\square$

Remark 7.7. Theorem 7.6 captures a counter-intuitive phenomenon: an observer can become topologically more complex through forgetting. The emergence of inaccessible repair cycles — holes in the admissibility manifold — represents a kind of structural complication that progressive erasure introduces even as the total admissible volume shrinks. This is the geometric analogue of the observation that partial memory loss can produce confabulatory behaviour: the system develops topological constraints it cannot see and cannot navigate around.

Remark 7.8. Ray Barman et al. [2] observe a sharp phase transition in H_0 and a peak in H_1 of semantic embedding spaces at a critical connectivity threshold, with H_1 reaching 534 ± 24 at the transition. This is an empirical instance of the Hole Creation Theorem: at the critical threshold, the removal of connectivity between semantic clusters creates non-contractible loops in the surviving space, exactly as Theorem 7.6 predicts. The persistence diagram of their embedding space is the empirical correlate of $PH_*(A)$ for a retrieval observer undergoing progressive interference-driven erasure.

Definition 7.9 (Persistent topological memory). A topological feature of $PH_*(A)$ *persists* if its lifespan $t_d - t_b$ in the persistence diagram exceeds a threshold $\tau > 0$. Persistent features are the structurally robust components of memory; ephemeral features are fragile repair cycles that erasure quickly destroys. The persistence diagram provides a complete summary of which repair structures survive progressive operator loss.

8. Biological Forgetting

The biological substrate for distinction-maintaining repair is the neural system. Repair operators include synaptic strengthening, protein synthesis, glial repair, and the broader repertoire of cellular maintenance processes. Erasure events include synaptic weakening, synaptic elimination, cell death, and axonal degeneration.

Example 8.1 (Synaptic decay and partial hysteresis). Long-term potentiation (LTP) is a repair operator that strengthens synaptic connections, increasing the probability of successful distinction restoration. Synaptic weakening below the LTP threshold is reversible: $0 < H_t(d) < +\infty$. Synaptic elimination by pruning or excitotoxicity is in general irreversible: $H_t(d) = +\infty$ if the cellular machin-

ery for synapse formation is also lost. The transition between these regimes is a biological instance of the boundary crossing described by Theorem 6.3.

Example 8.2 (Neural injury and admissibility contraction). Focal cortical lesions remove repair operators acting on specific regions of the substrate. By the Admissibility Contraction Theorem, the admissibility manifold of the injured system is a strict subset of the intact system’s manifold. The deformation tensor E_t is non-trivial near the lesion boundary: distinctions that were easily maintained before the lesion become increasingly costly as they approach the injured region. This is the geometric correlate of the penumbra effect in stroke: the region surrounding the infarct in which repair is expensive but not yet impossible.

Example 8.3 (Developmental pruning and homology change). Synaptic pruning during development eliminates a substantial fraction of early repair operators. If pruned synapses were cut-points of the admissibility manifold, Theorem 7.5 predicts fragmentation of the repair domain. Conversely, if pruning eliminates redundant paths while preserving connectivity, $H_0(A_t)$ is unchanged and only higher-order homology is affected. The distinction between pathological and healthy pruning may correspond precisely to whether pruning changes H_0 (harmful) or only higher homology (reorganising without fragmenting).

9. Institutional Forgetting

Institutions are observers whose repair repertoires consist of codified procedures, trained personnel, and maintained infrastructure. Erasure events include staff attrition, procedural reform, budget cuts, and physical destruction of facilities.

Example 9.1 (Archive destruction). A physical archive is a repair operator: it restores distinctions between historical states by maintaining retrieval paths from present queries to past records. Destruction of an archive removes those operators. By the Permanent Forgetting Theorem, distinctions whose only repair paths passed through the destroyed archive acquire $H_t(d) = +\infty$: they cannot be restored at any finite cost from the reduced repertoire. The distinctions do not become false; they become unrestorable. This is the precise sense in which archive destruction is epistemically irreversible.

Example 9.2 (Language death). A living language is a repair repertoire for semantic and phonological distinctions. Language death is an erasure path: as

speakers die or shift to other languages, the repair operators (speakers capable of producing and recognising the language’s distinctions) are progressively removed. The deformation tensor E_t is non-trivial near the distinctions specific to the dying language: they become increasingly costly to maintain as the speaker population falls. Total collapse of the speaker population corresponds to $\text{Rest}(\mathcal{R}_t) = \emptyset$: topological total forgetting. Language revitalisation is the re-acquisition of repair operators, with a hysteresis cost determined by how far along the erasure path the language has progressed.

Example 9.3 (Institutional collapse). The collapse of a legal or governmental institution removes procedural repair operators. The Paradigm Shift Theorem of [1] showed that non-trivial repertoire change produces incomparable restorable sets. Institutional collapse is the limiting case: the entire repertoire is lost, and the new institution that replaces it typically has a different restorable set, not a subset. Reconstruction of collapsed institutions is path-dependent: different reconstruction paths through the sub-repertoire lattice produce different admissibility geometries even when the nominal repertoire is nominally restored. Meadows [12] identifies *eroding goals* and *policy resistance* as characteristic system traps: when a system loses the feedback loops that maintain a standard, the standard itself erodes rather than the gap between standard and performance being corrected. In repair-theoretic terms, eroding goals correspond to a slow erasure path in which the operators that enforce a distinction are progressively weakened before being eliminated, with the distinction boundary migrating inward — precisely the moving-front dynamics of Theorem 6.8 — before the distinction becomes permanently irrestorable.

10. Scientific Forgetting

Scientific knowledge is maintained by a collective repair repertoire: experimental techniques, trained practitioners, instrumentation, and institutional infrastructure. Scientific forgetting is the loss of repair operators from this collective repertoire.

Example 10.1 (Loss of experimental technique). Certain experimental techniques become inaccessible when the practitioners who developed them retire without training successors. The technique is an operator $r \in \mathcal{R}_0$; its loss is an erasure event. Distinctions whose restoration depended exclusively on that technique

acquire $H_t(d) = +\infty$ if the technique cannot be reconstructed from surviving operators. This is institutionally invisible: the distinctions remain nominally part of the scientific record but are practically unrestorable.

Example 10.2 (Paradigm transitions as structured erasure). The Paradigm Shift Theorem of [1] characterised paradigm transitions as producing incomparable restorable sets. Under the erasure framework, a paradigm transition is an erasure path followed by a repertoire acquisition path: old operators are removed, new ones are added. The net effect on the deformation tensor depends on whether the new operators can restore the distinctions made irrestorable by the old operators' removal. If not, the transition involves permanent scientific forgetting — the Kuhnian claim that paradigm shifts make some distinctions permanently invisible — formalised as $H_T(d) = +\infty$ for distinctions in $\text{Rest}(\mathcal{R}_0) \setminus \text{Rest}(\mathcal{R}_T)$.

Example 10.3 (Replication crisis as repair failure). The replication crisis in empirical sciences can be recast as a collective repair failure: experimental results that were nominally in $\text{Rest}(\mathcal{R}_0)$ are found, upon attempted restoration under independent operators, to have $C_R(d, \mathcal{R}_0) = +\infty$ — they were never genuinely restorable, only apparently so. The crisis is the discovery that the admissibility boundary was mislocated: distinctions believed to be in the interior of $\mathcal{A}_0$ were in fact on or outside it.

11. Consequences, Open Problems, and Future Directions

The framework developed here yields a unified account of phenomena that are conventionally treated as belonging to different domains.

Memory is geometry. The metrised repair space $(\text{Rest}(\mathcal{R}), d_R)$ is the formal object corresponding to what an observer remembers. Memory loss is not subtraction of stored items but deformation of this metric space, measured by the tensor E_t .

The right objects are invariants, not components. The deformation tensor E_t is a symmetric $(0, 2)$ -tensor; its components in any particular basis carry no direct physical meaning. The meaningful objects are invariants: the operator norm $|E_t|$ measures total deformation; the eigenvalues of E_t (relative to g_0) identify principal directions of forgetting; the trace $\text{tr}(E_t)$ measures mean distortion; the

determinant measures volume change in the surviving repair domain. This is precisely how strain tensors are interpreted in continuum mechanics: not component by component, but through invariant quantities such as principal strains, volumetric strain, and deviatoric strain. A fading memory corresponds not to a particular tensor component becoming large but to a specific invariant — likely $|E_t|$ or a principal eigenvalue — diverging in the direction associated with the distinction being lost.

Forgetting is irreversible in the generic case. The hysteresis functional $H_t(d)$ is non-decreasing along any erasure path. For biological and institutional observers, the operators most likely to be lost — cell death, archive destruction, speaker extinction — are precisely those for which $H_t(d) = +\infty$. Reversible forgetting is the special case, not the default.

Restoration is path-dependent. Two observers that have undergone different erasure paths to the same reduced repertoire may have different restoration costs, because intermediate states of the admissibility manifold affect what residual structure is available for reconstruction. This connects the framework to the broader distinction-history program: what matters is not the current state but the trajectory.

Forgetting has two independent modes. Metric deformation ($|E_t| \rightarrow \infty$) and topological change ($H_*(A_t)$ changing) are independent phenomena. The full geometry of forgetting requires both: metric deformation describes how costly access becomes; topological change describes which distinctions lose mutual accessibility entirely. A storage theory of memory can represent neither.

Open Problems

1. **Boundary migration speed.** The migration measure $V(t) = \inf_{x \in \partial A_t} \inf_{y \in \partial A_0} d_R^{(0)}(x, y)$ is defined but its time derivative is not established. Under what regularity conditions on the erasure path and the admissibility boundary is $V(t)$ differentiable? When it is, $\dot{V}(t)$ is the migration speed — a scalar quantifying how fast the irrestorability front advances. Is there an equation of motion for the front analogous to the eikonal equation in geometric optics?
2. **Rate of deformation.** How does E_t evolve along an erasure path? Is there a flow equation $\partial_t E_t = F(E_t, g_0, \mathcal{R}_t)$? The analogy with Ricci flow is sug-

gestive: curvature-driven deformation might model how metric singularities propagate through the repair domain after a localised erasure event. Ricci flow develops singularities in finite time, which would correspond to the onset of permanent forgetting.

3. **Critical repair operators.** The Fragmentation Theorem identifies cut-points as structurally critical distinctions; the Hole Creation Theorem identifies contractible removable regions. Characterise the full set of *critical repair operators*: those whose removal changes $H_*(A_t)$. Are there operators that stabilise the manifold topology against the removal of other operators? If so, they are candidates for targeted preservation in biological, archival, and linguistic contexts.
4. **Minimal restoration.** Given a reduced repertoire $\mathcal{R}_t$ and a target distinction set $D \subseteq \text{Rest}(\mathcal{R}_0) \setminus \text{Rest}(\mathcal{R}_t)$, what is the minimal operator set that must be re-acquired to restore D ? This is an inverse problem on the sub-repertoire lattice $\mathcal{L}(\mathcal{R}_0)$. The orbit structure of the erasure monoid (Section 3) constrains the answer: not all paths from $\mathcal{R}_t$ to a repertoire restoring D have equal cost.
5. **Erasure and the interference matrix.** How does the interference matrix $\mathbf{I}$ of [1] evolve under erasure? Does operator loss preferentially increase positive eigenvalues (more repair conflict among surviving distinctions) or decrease negative eigenvalues (less synergy)? The effective dimensionality concentration found by Ray Barman et al. [2] — all tested embedding models concentrating variance into roughly 16 effective dimensions regardless of nominal dimensionality — suggests that biological and artificial memory systems operate in a regime where a small number of operators account for most repair capacity, making them catastrophically vulnerable to targeted erasure. A prediction: erasure of the operators corresponding to the top eigenvectors of $\mathbf{I}$ should produce disproportionate fragmentation relative to erasure of random operators.
6. **Optimal erasure paths.** On $\mathcal{L}(\mathcal{R}_0)$, characterise the erasure paths that minimise total deformation $\int_0^T |E_t| dt$ subject to a fixed endpoint $\mathcal{R}_T$. Such paths represent maximally graceful forgetting. The dual problem — paths that maximise $\int_0^T |E_t| dt$ — characterises maximally disruptive forgetting, relevant to adversarial institutional erasure.

Future Directions

The following questions lie beyond the scope of this paper but are natural extensions of its framework.

Stratified admissibility manifolds. The current framework assumes A_t is a topological manifold. The Fragmentation and Hole Creation Theorems predict, however, that progressive erasure can produce non-manifold spaces: boundaries may become fractal, strata may appear at different dimensions, and the singular set of A_t may itself have non-trivial topology. The natural generalisation is to work in the category of *stratified spaces* [5], where the metric tensor lives on each stratum and the homology groups are computed globally. This would accommodate the full range of singularities that erasure generates without requiring smoothness assumptions that cannot be established in general.

Thermodynamics of erasure. The framework uses repair cost as a primitive but does not specify its physical nature. Landauer’s principle [11] establishes that the erasure of one bit of information requires at least $kT \ln 2$ of work. If distinctions are identified with bits and erasure events are identified with logical irreversibility, the hysteresis functional $H_t(d)$ should satisfy a Landauer-type lower bound: $C_R^{\text{destroy}}(d) \geq kT \ln 2$ for any distinction whose erasure is informationally irreversible. Whether the deformation tensor E_t is conjugate to a stress-energy tensor in the physical sense — whether there is a genuine variational principle underlying erasure deformation — is an open question connecting this framework to the thermodynamics of computation and the physics of information.

Continuous operator weights. The current framework treats repair operators as present or absent: $r \in \mathcal{R}$ or $r \notin \mathcal{R}$. In biological systems, synaptic connections are weakened continuously before elimination; in institutional systems, procedures are degraded incrementally before abandonment. A natural extension defines operator weights $w_r \in [0, 1]$ and replaces the discrete lattice $\mathcal{L}(\mathcal{R}_0)$ with a continuous space of weighted repertoires. The deformation tensor E_t would then vary continuously with the weights, and its singularities would correspond to weight thresholds at which distinctions cross from restorable to irreparable. Partial hysteresis ($0 < H_t(d) < +\infty$) is precisely the regime this extension would formalise rigorously.

Observer identity deformation. The current paper assumes observer persis-

tence: the observer before and after the erasure path is the same observer with reduced capacity. For sufficiently severe forgetting, this assumption fails. A stroke patient with extensive cortical damage, a language community reduced to one elderly speaker, or an institution that has lost all organisational memory may constitute a different observer rather than a depleted version of the same one. The natural question is whether observer identity is itself definable as a path-connected component of the category of admissibility manifolds: two observers are the same if there exists an erasure path connecting their repertoires within a single connected component; they are different if no such path exists. This would make observer identity a topological invariant of the admissibility geometry, and observer deformation — the passage between observer-identity classes — a phenomenon requiring the full categorical machinery to describe. It is probably the deepest question to which the current framework points.

Erasure is not information loss. It is the deformation of a world. What is forgotten is not gone; it has moved to a boundary where repair costs diverge, where the metric of what remains is singular, and where the topology of the admissible has been irreversibly changed. The observer that forgets is not smaller. It is differently shaped.

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